

ELECTRONIC HEALTH RECORDS IN THE MODERN WORLD

Electronic Health Records in the Modern World

- In the last 20 years, Electronic Medical/Health Records have been implemented in many countries
- Mostly starting with hospitals
- Primary Care usually lagging behind
- Not to speak about ...
 - independent physiotherapists
 - independent psychotherapists
- Implementation of EHRs strongly depends on the health care system:
 - centralized healthcare system
 - example: United Kingdom - NHS
 - semi-centralized healthcare system
 - example: Austria - each province can have different systems
 - decentralized healthcare system
 - example: USA

Electronic Health Records and Interoperability

- Most hospitals nowadays have an EHR or EMR
- Even developing countries can implement EHRs there is a lot of open source EHR systems
 - example: OpenEMR: <https://www.open-emr.org>
- Major problem however is interoperability!
 - structural: exchange formats
 - semantic: coding systems
- Semantic interoperability is harder than structural interoperability ...
 - Fortunately, the whole world is standardizing more and more to:
 - LOINC for tests
 - SNOMED-CT
 - UCUM for Units of Measure
 - ATC, ICD, ...
 - Except for ... clinical research (CDISC, FDA, ...)

- When all hospitals use the same system (example: Slovenia), do we still need an exchange format?
- Also such system (e.g. OpenEHR) implementations can be decentralized => exchange format still necessary, but can be part of the system
- What with patient access, especially aggregation of data? Example: patient wants to see all blood pressure measurements of last years, independent of where these have been taken
- When different health care providers use different systems use 5 different systems, standardized exchange standards will be necessary

- We need to clearly distinguish between EHRs as a system, and standardized exchange of EHRs
- KAGES and use different internal systems. Can they exchange EHRs?
- Slovenia uses OpenEHR in all hospitals All patients have full access to their EHRs Is there an exchange standard?
- HL7-FHIR is an exchange system for EHRs but doesn't impose how the data is captured or stored

- Healthcare is organized by province (Bundesland).
- Information exchange between primary care and is limited (based on "letters")
 - if you PCP is measuring your blood pressure, can you later find it back?
- Hospital information systems differ:
 - between provinces for province-owned hospitals
 - private hospitals have other systems
 - what about dentists, physiotherapists, psychotherapists
 - what about rehabilitation centers?

- 2019: mountain bike accident => UKH Graz
 - only lab result in ELGA is blood group type
 - discharge letter is about 1 page
- 2020: 3 weeks in rehabilitation center (broken upper arm)
 - No entry at all in ELGA

ion e-Impfpass ELGA-GDA ELGA-Teilnahme Protokoll

Keine neuen e-Befunde seit dem letzten Logout

Aktive Filter: Gesamter Zeitraum

Befundtyp Titel	Datum	GDA	
Entlassungsbrief Entlassungsbrief Arzt	16.01.2019	Unfallkrankenhaus Graz	▼
Befund bildgebende Diagnostik CT-Befund	16.01.2019	Unfallkrankenhaus Graz	▼
Laborbefund Laborbefund	15.01.2019	Unfallkrankenhaus Graz	▼
Laborbefund Laborbefund	15.01.2019	Unfallkrankenhaus Graz	▼

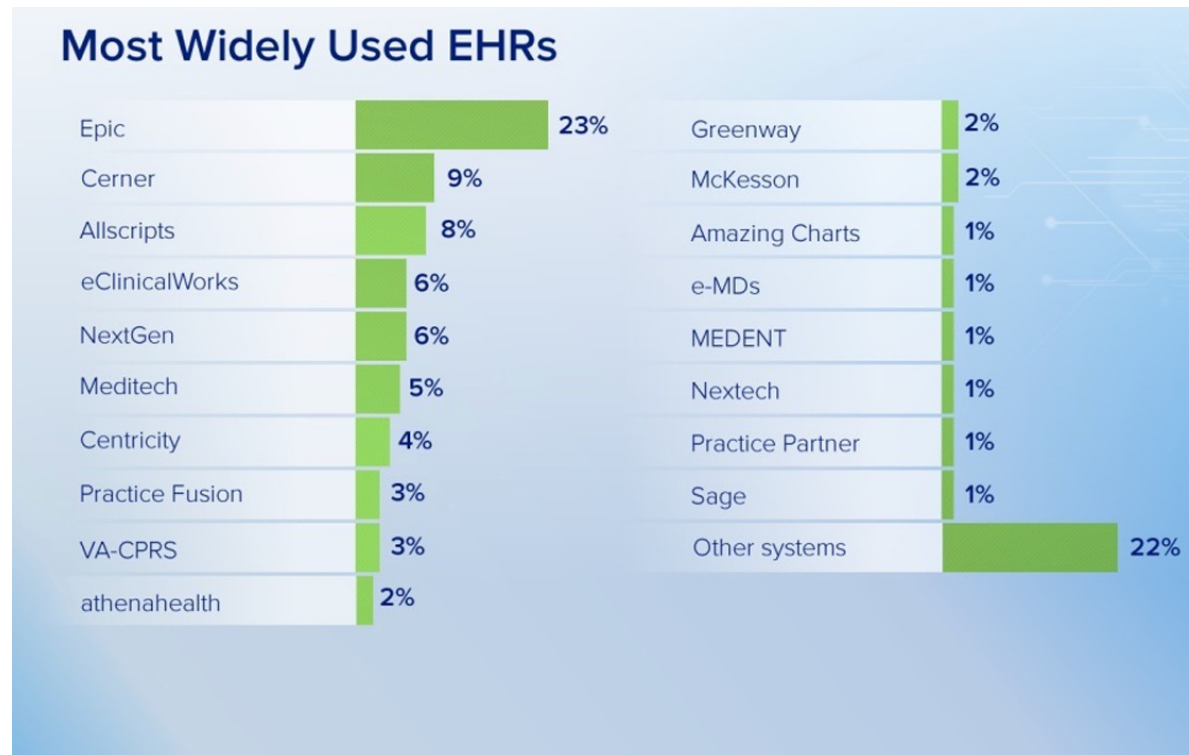
- e-Medication added 2017-2018
- e-vaccination pass added 2021
 - what about COVID-19 test results?

- Based on HL7-v3 CDA (XML) documents
 - Implementation Guide for each type of document
- Decentralized repositories
- Central system for access authorization and document lookup using IHE profiles (XDS, ...)

- USA: CCR/CCD (Continuity of Care Record/Document)
- FHIR
 - Overview
 - API in detail
 - FHIRPath
 - HAPI-FHIR server and Java libraries
 - How to implement FHIR in Java Using open test servers
 - CRUD: creating and adding patients
- OpenEHR architecture

THE US CONTINUITY OF CARE RECORD

- Hospitals usually do have an EHR/EMR systems for a longer time
- A few large IT companies dominate the market
- Usually, patients have access to the hospital EHR through a portal
- These EHRs systems are "in situ" not interoperable

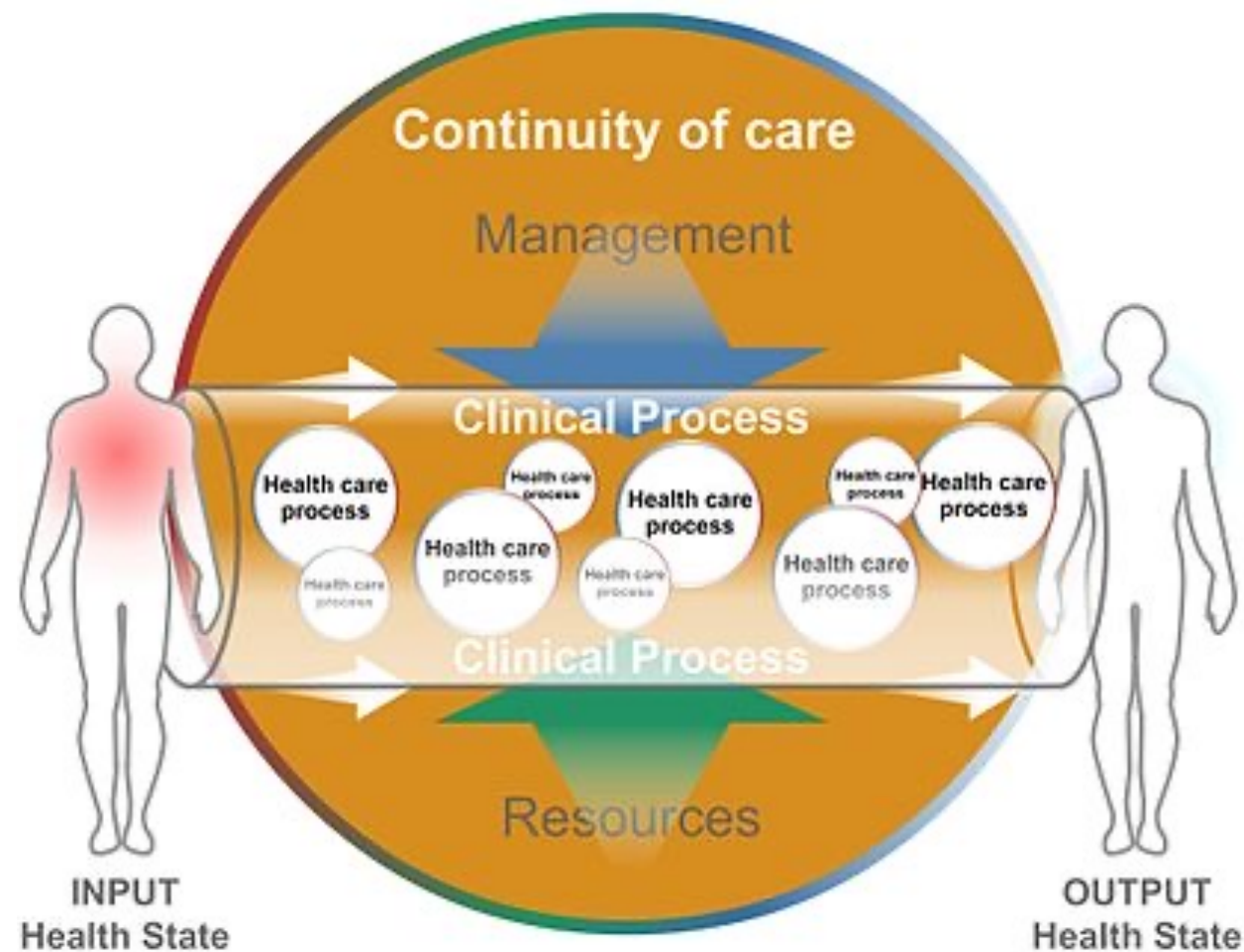


- The Meaningful Use Program (started 2009, American Reinvestment Recovery Act) had the aim to promote the use of (exchangeable) and EHRs in healthcare
- The CCR/CCD (based on HL7-v3 CDA) was introduced as a **minimum** requirement for EHR exchange
- CCR/CCD are limited to the "continuity of care" use case

- "Continuity of Care Record"
 - Developed for the USA
 - But also used in other countries
- Somewhat like "ELGA Entlassungsbrief", but ... more
- A summary, not a full health record
- Predefined content and sections

Continuum of Care



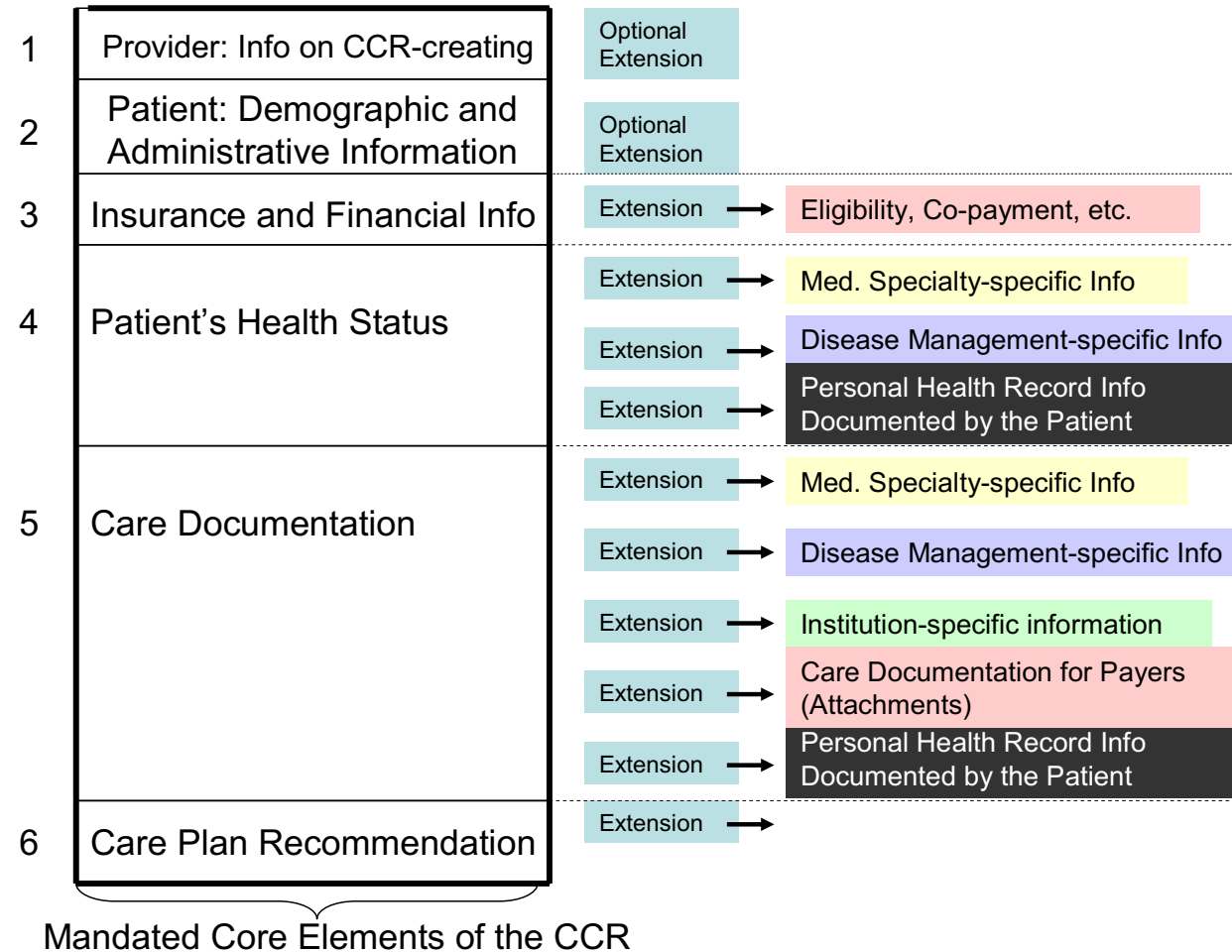


- Document Identifying information
 - From / to / reason for referral/transfer
- Patient identifying information
- Insurance and financial information
- Health status of Patient
 - See next slides
- Case documentation
 - See next slides
- Care plan Recommendation(text)

- Problems / Diagnoses / Conditions
- Adverse reactions & alerts
- Current medications
- Immunizations
- Vital signs
- Lab results
- Procedures & assessments

- Med. specialty-specific information
- Disease-management-specific information
- Institution-specific information
- Including information about previous encounters (date + time, ...)
- Care documentation for "payers" (insurance)
 - Through attachments
- Personal Health Record information
 - Documented by the patient

Conceptual Model of the CCR



- In the USA it is not allowed to use the Social Security Number for healthcare identification see e.g

<http://library.ahima.org/doc?oid=104465#.V9ZggSu4J64>

In recent years, it has become increasingly easy to use SSNs for fraudulent purposes, partly because of extensive illegal sales of SSNs. "As long as criminals can buy a list of names and SSNs through an Internet auction, we will continue to be plagued by the consequences," stated Patrick O'Carroll, inspector general of the Social Security Administration.²

Exacerbating the problem of using the SSN as a patient identifier is the fact that patient registration staff typically ask for the SSN verbally and, if the staff do request the card, it contains no picture or biometric identifiers. This makes it essentially impossible to ascertain if the person using the card is the one to whom the card was issued.

- Identification based on name and birth date is very difficult

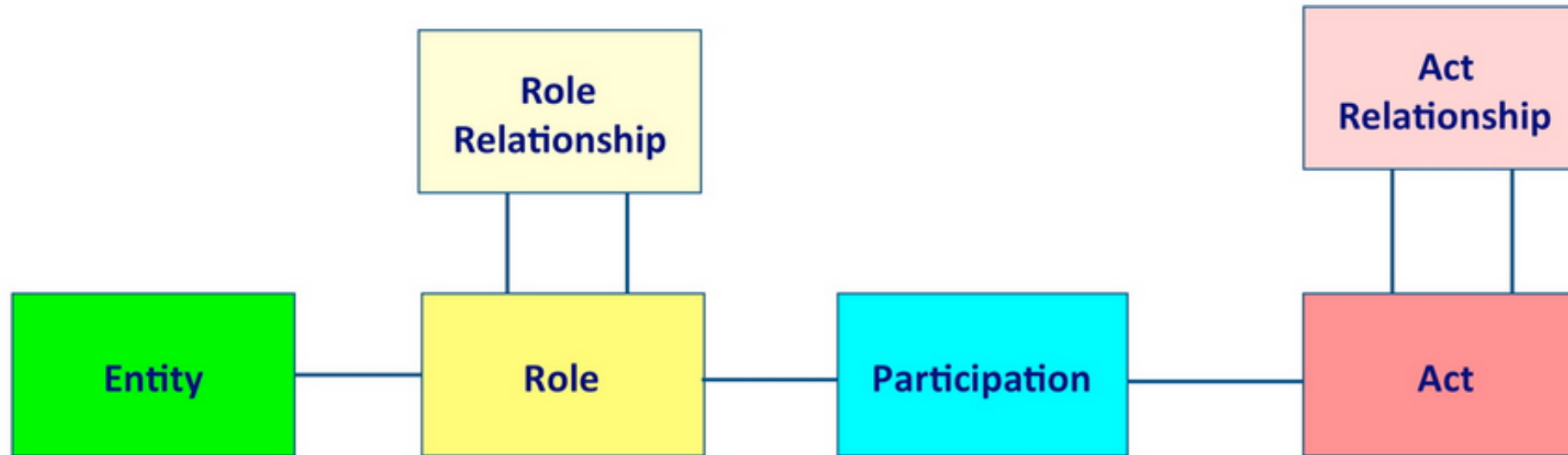
“I have a Maria Gonzales in my emergency room. She said she had a car accident three years ago in Chicago and had some stuff taken out, but she doesn’t remember what. The hospital in Chicago has hundreds of Maria Gonzales’s in their system. I do not want to act on the information of the wrong Maria Gonzales.”

- CCR does not impose any identification scheme

- ASTM E2369 - 12
 - XML implementation by ASTM
 - Described by an XML-Schema
 - See e.g. <http://www.veoh.com/watch/v14141513WQRzgizc> (movie)
 - Being faded out - hard to find examples ...

- CCD: Continuity of Care Document
 - HL7-CDA implementation

- CDA is based on RIM - Reference Information Model



- Each XML-element in a CDA document must correspond to a RIM class or class attribute

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- edited with XMLSpy v2010 (http://www.altova.com) by Len Gallagher (NIST) -->
<!--Sample XML file generated by XMLSpy v2007 sp2 (http://www.altova.com)-->
<ClinicalDocument xmlns="urn:hl7-org:v3" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="urn:hl7-org:v3 http://xre:
  <realmCode code="US"/>
  <typeId root="2.16.840.1.113883.1.3" extension="POCD_HD000040"/>
  <templateId root="2.16.840.1.113883.3.27.1776" assigningAuthorityName="CDA/R2"/>
  <templateId root="2.16.840.1.113883.10.20.3" assigningAuthorityName="HL7/CDT Header"/>
  <templateId root="1.3.6.1.4.1.19376.1.5.3.1.1.1" assigningAuthorityName="IHE/PCC"/>
  <templateId root="2.16.840.1.113883.3.88.11.32.1" assigningAuthorityName="HITSP/C32"/>
  <id root="2.16.840.1.113883.3.72" extension="HITSP_C32v2.5_Rev3_11Sections_NoEntries_EntryErrorsOnly" assigningAuthorityName="NIST Health
  <code code="34133-9" displayName="Summarization of episode note" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"/>
  <title/>
  <effectiveTime value="20091110130945"/>
  <confidentialityCode/>
  <languageCode code="nn-CC"/>
  <recordTarget>
    <patientRole>
      <id/>
      <addr use="HP"/>
      <telecom/>
      <patient>
        <name>
          <administrativeGenderCode code="F" displayName="Female" codeSystem="2.16.840.1.113883.5.1" codeSystemName="HL7 Administrative
          <birthTime value="19840704"/>
          <maritalStatusCode code="S" displayName="Single" codeSystem="2.16.840.1.113883.5.2" codeSystemName="HL7 Marital status"/>
          <languageCommunication>
            <templateId root="2.16.840.1.113883.3.88.11.83.2" assigningAuthorityName="HITSP/C83"/>
            <templateId root="1.3.6.1.4.1.19376.1.5.3.1.2.1" assigningAuthorityName="IHE/PCC"/>
            <languageCode code="en-US"/>
          </languageCommunication>
        </patient>
      </patientRole>
    </recordTarget>
  </author>
```

- As CCR is in XML (either ASTM or CDA), it can easily be transformed using stylesheets (XSLT) into:
 - HTML → web view
 - PDF → "paper-like" view
 - ...
 - Even into Word documents!

- MU-Stage 1:
 - HL7-CCD and ASTM-CCR were both selected as acceptable extract formats for clinical care summaries
 - EHRs must be able to generate a CCD (or equivalent CCR) that has the sections of allergies, medications, problems, and laboratory results, in addition to patient header information
 - Most Electronic Health Record vendors have adopted the HL7-CCD rather than the ASTM-CCR since it is a newer format

See: https://en.wikipedia.org/wiki/Continuity_of_Care_Document

- MU-Stage 2:
 - CCD, but not the CCR, was included as part of the standard for clinical document exchange
 - HL7 developed the "**consolidated CDA**" (C-CDA)
 - C-CDA includes nine document types, one of which is an updated version of the CCD
 - MU-2 requires that healthcare providers use C-CDA document exchange regularly in care transitions and the CCD has been identified as the most appropriate document for this purpose

C-CDA Implementation Guide at:

http://www.hl7.org/implement/standards/product_brief.cfm?product_id=379

The Consolidated CDA implementation guide defines nine different types of commonly used CDA documents, including:

- Continuity of Care Document
- Consultation Notes
- Discharge Summary
- Imaging Integration, and DICOM Diagnostic Imaging Reports
- History and Physical
- Operative Note
- Progress Note
- Procedure Note
- Unstructured Documents

Source: <https://corepointhealth.com/health-standards-news-ccd-and-consolidated-cda>

- How the Continuity of Care Document Can Advance Medical Research and Public Health
- The Clinical Document Architecture and the Continuity of Care Record: A Critical Analysis
- Continuity of Care Document for Hospital Management Systems: an Implementation Perspective